

1. Write an algorithm and draw a flowchart to check whether a given number is divisible by both 3 and 5, only by 3, only by 5, or by neither.
2. A movie ticket system charges Tk. 300 for adults and Tk. 200 for children. If the showtime is after 6 PM, a 10% evening surcharge applies. Write an algorithm and draw a flowchart to calculate the total cost for a family.
3. A student's performance in a course is evaluated using class tests, assignments, and attendance. If the average of tests and assignments is 80 or above and attendance is 75% or more, the student is declared "Excellent." If the average is between 60 and 79 but attendance is at least 60%, the grade is "Good." Otherwise, the student is marked "Needs Improvement." Write an algorithm and draw a flowchart to determine the final remark.
4. A temperature monitoring system reads the current temperature and humidity level of a room. If the temperature exceeds 30°C and humidity is below 40%, it should print "Hot and Dry." If the temperature is above 30°C but humidity is above 70%, it should print "Hot and Humid." If temperature is below 20°C and humidity is above 70%, it should display "Cold and Damp." For all other cases, it should print "Comfortable." Write an algorithm and draw a flowchart for this.
5. A shop offers discounts based on total purchase amount and membership type. If the customer is a premium member and spends more than 5000 BDT, they get a 20% discount. If they are a regular member spending more than 5000 BDT, they get a 10% discount. If any customer spends less than 1000 BDT, no discount is applied. Otherwise, a flat 5% discount is given. Write an algorithm and draw a flowchart to calculate the final amount after discount.
6. An electricity bill is calculated based on units consumed. If the units are less than 100, the rate is 5 BDT per unit. For units between 100 and 300, the rate is 8 BDT per unit. For units above 300, the rate is 10 BDT per unit. Additionally, if the total bill exceeds 2000 BDT, a 15% surcharge is added. Write an algorithm and draw a flowchart to compute the total payable amount.
7. A mobile operator charges differently depending on call duration and time of day. If the call lasts less than 5 minutes, a flat rate of 1.5 BDT per minute applies. If it lasts longer, the rate becomes 1.2 BDT per minute. However, if the call occurs during nighttime (after 10 PM or before 6 AM), a 25% discount is applied to the total charge. Write an algorithm and draw a flowchart to compute the final cost.
8. A company's annual bonus policy depends on both years of service and performance rating. If an employee has worked for more than 10 years and has a rating of 4 or higher, they receive a 20% bonus. If the service is between 5 and 10 years and rating is at least 3, they receive 10%. If either the service is below 5 years or rating below 3, they get no bonus. Write an algorithm and draw a flowchart to determine the bonus amount.

Write an algorithm and draw a flowchart to calculate sum of the each of the following series –

- $1 + 2 + 3 + \dots + n$
- $1^2 + 2^2 + 3^2 + \dots + n^2$
- $1^3 + 2^3 + 3^3 + \dots + n^3$
- $2 + 4 + 6 + 8 + \dots + n$
- $1 + 3 + 5 + 7 + \dots + n$
- $1 - 2 + 3 - 4 + 5 - 6 + \dots + n$
- $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots + n^2$
- $1 + 2 + 4 + 8 + 16 + \dots + 2^n$
- $1 + 5 + 9 + 13 + \dots + (4n - 3)$
- $5 + 10 + 20 + 40 + \dots + n$
- $1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n \times (n + 1)$
- $1^1 + 2^2 + 3^3 + \dots + n^n$
- $1 + 1/2 + 1/3 + 1/4 + \dots + 1/n$
- $2 + 6 + 12 + 20 + 30 + \dots + n(n + 1)$
- $1 + 4 + 9 + 16 + 25 + \dots + n^2$
- $1 + 3^2 + 5^3 + 7^4 + \dots + (2n - 1)^n$